

CL-1002
Programming Fundamentals
Lab # 11

Objectives:

- Nested Loops

Note: Carefully read the following instructions (*Each instruction contains a weightage*)

1. First think about statement problems and then write your program.
2. Write Program in C compiler/IDE and save source file **for each program**.
3. **Do not copy from any source otherwise you will be penalized with negative marks.**
4. Complete your lab **within given Time Slot**.
5. Add your source code in this word document + Make one ZIP file of your all source codes.
6. Please submit your **Both files** with this naming convention ROLLNO_SECTION_LABNO.
7. Submit your lab on Google Classroom.

Problem: 1

Pascal's Triangle

Sample Example:

- Input: n = 5
- Output:
- 1
- 1 1
- 1 2 1
- 1 3 3 1
- 1 4 6 4 1

Problem: 2

- **Problem Statement:** Write a C++ program to generate 'n' random numbers between a specified range (min and max), such that no number is repeated.
- **Sample Example:**
 - Input: n = 5, min = 1, max = 10
 - Output: Generated random numbers: 3 9 1 7 4

Problem: 3

- Write a C++ program to find the length of the longest palindromic subsequence in a given string of characters. The program should output both the length and the subsequence.

- **Sample Example:**

- Input: "character"
- Output: Length of the longest palindromic subsequence: 5 Longest palindromic subsequence: "carac"

Problem: 4

- Given an array of 'n' positive integers, write a C++ program to find the difference between the sums of two subsets of the array, such that the absolute difference is minimized.

- **Sample Example:**

- Input: Array = {3, 1, 4, 2, 2}
- Output: Minimum subset difference: 0 Subsets with the minimum difference: {3, 4} and {1, 2, 2}

Problem: 5

- Write a C++ program to generate all possible permutations of a given set of 'n' distinct integers. The program should print each permutation.

- **Sample Example:**

- Input: Set = {1, 2, 3}
- Output:
- Permutations:
- 1 2 3
- 1 3 2
- 2 1 3
- 2 3 1
- 3 1 2
- 3 2 1

Problem 6:

- Write a C++ program to generate and print all Gray codes of a given 'n' bits using nested loops. Gray codes are binary sequences where adjacent numbers differ by only one bit.

- **Sample Example:**

- Input: n = 3
- Output:

000 001 011 010 110 111 101 100

Problem: 7

- Write a C++ program to find the sum of all divisors of a given positive integer 'n'. The divisors of 'n' are the positive integers that evenly divide 'n'. Implement this without using arrays or data structures.

- **Sample Example:**

- Input: n = 12
- Output: Sum of divisors of 12: 28 (1 + 2 + 3 + 4 + 6 + 12)

Problem: 8

- Write a C++ program to check if a given integer is a palindrome. A palindrome number reads the same backward as forward (e.g., 121, 1331). Implement this without converting the integer to a string.

- **Sample Example:**

- Input: n = 12321
- Output: 12321 is a palindrome.